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Project Title:	Systems Biology Model of Interactions between Tissue Growth Factors and DNA Damage Pathways: Low Dose Response and Cross-Talk in TGFBeta and ATM Signaling		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	89154-9900	**Congressional District:**	1
Comments:	NOTE: Formerly at Lawrence Berkeley National Laboratory until fall 2017.		
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Flight Program:			
Flight Assignment:	NOTE: New end date of 12/31/2014 (previously 1/31/2014) per PI and LBL information (Ed., 3/20/14)		
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Task Description:	The etiology of radiation carcinogenesis has been described in terms of aberrant changes that span several levels of biological organization. Growth factors regulate many important cellular and tissue functions including apoptosis, differentiation, and proliferation. A variety of genetic and epigenetic changes of growth factors have been shown to contribute to cancer initiation and progression. It is known that cellular and tissue damage to ionizing radiation is in part initiated by the production of reactive oxygen species, which can activate cytokine signaling, and the DNA damage response pathways, most notably the ATM signaling pathway. Recently the transforming growth factor beta pathway has been shown to regulate or directly interact with the ATM pathway in the response to radiation. The relevance of this interaction with the ATM pathway is not known although p53 becomes phosphorylated and DNA damage responses are involved. However, growth factor interactions with DNA damage responses have not been elucidated particularly at low doses and further characterization of their relationship to cancer processes is warranted. Our goal will be to use a systems biology approach to mathematically and experimentally describe the low dose responses and cross-talk between the ATM and TGF beta pathways initiated by low and high LET radiation. We will characterize ATM and TGF beta signaling in epithelial and fibroblast cells using 2D models and compare these data to signaling patterns in physiologically relevant 3D organotypic cell culture models to elucidate possible cell type and context specific differences in inter-cellular communication. We will investigate the roles of the Smad and Activating transcription factor 2 (AFT2) proteins as the potential major contributors to crosstalk between the TGF beta and ATM pathways, and links to cell cycle control and/or the DNA damage response, and potential differences in their responses at low and high doses. We have developed various experimental approaches to apply to these problems, including confocal microscopy and flow cytometry, to detail changes at low dose/dose-rate in order to understand individual cell responses, and will establish our mathematical models based on the experimental findings resulting from change in DNA repair, apoptosis, and proliferation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Changes in growth factor signaling due to either genetic and/or epigenetic alterations have been shown to play a role in cancer initiation and progression. Transforming growth factor beta (TGFß) is rapidly and persistently activated in response to DNA damage. It has an important role in controlling cellular proliferation, and differentiation among other cellular functions. Cellular and tissue damage following ionizing radiation is in part initiated by the production of reactive oxygen species (ROS) that activate cytokine signaling, and the DNA damage response pathways, including the ATM signaling pathway. It has been proposed that radiation induced stimulation of ATM and its downstream components involves activation of TGFß-1 and may result from TGFß-1-receptor I-Smad signaling. It was suggested that perhaps TGFß signaling primes cells to respond to DNA damage either by helping to recruit ATM to the site of DNA damage, or by activating ATM. Further work confirmed that TGFß-1 had a role in stimulating ATM- and p53 following radiation, but revealed that rapid ATM stimulation was not dependent on TGFß-1-receptor I-Smad signaling. Thus the relationship between TGFß and ATM pathways is complex and the cross-talk between these pathways may differ dependent on the dose or radiation quality of the exposure. Dissecting the crosstalk between these pathways is the primary focus of our investigation. As the human esophagus is an organ sensitive to radiation exposure, we chose the human esophageal cell model as a platform to dissect the differences between cellular (2D) and 3D tissue responses to radiation. Experiments were designed to characterize ATM and TGFß signaling, as well as the interrelationship of each pathway's signaling by various means. Responses were monitored both in keratinocytes and fibroblast cells of esophageal origin, to better understand cell type differences in response. How differences in growth conditions may uniquely influence response was also studied by comparing signaling patterns observed in 2D monolayers to a more physiologically relevant 3D organotypic cell culture model. Studies also investigated the roles of the Smad and Activating transcription factor 2 (AFT2) proteins as potential major nodes for crosstalk between the TGFß and ATM pathways. In these studies we have investigated how cross-talk may influence cell cycle control and/or the DNA damage response, and investigated potential differences in responses at low versus high doses. Both confocal microscopy and flow cytometry are used to study the responses, to detail changes at low dose to understand individual cell responses. Our goal is to use a systems biology approach to mathematically and experimentally describe the low dose responses and cross-talk between the ATM and TGFß pathways initiated by low versus high LET radiation. Summary of research accomplishments: In these studies we have focused on defining the sensitivity of the human esophagus to radiation exposure and how radiation quality may impact that sensitivity. In addition, experiments were designed to characterize ATM and TGFß signaling, and the interrelationship of each pathway, and how dose and radiation quality may differentially effect this signaling. Responses were monitored both in epithelial and fibroblast esophageal cells, to better understand cell type differences. Responses were also monitored both in 2D monolayers as well as using a more physiologically relevant 3D organotypic cell culture model. Major nodes used to investigate for crosstalk between the TGFß and ATM pathways included the Smad and Activating transcription factor 2 (ATF2) proteins. Immunofluorescent studies using esophageal epithelial cells (EPC hTERT), plus and minus TGFß revealed that the addition of TGFß increased the intensity of radiation-induced pATF2 foci, a phosphorylation mediated by ATM. This suggested that there was some degree of cross-talk between ATM and TGFß in our cellular system, with TGFß perhaps amplifying the ATM mediated response post radiation. The ATM mediated specificity of this response was noted by the elimination of the majority of the ATF2 phospho-foci with ATM inhibitor addition. 3D organotypic studies, immunofluorescent studies were also performed to define if differences could be noted in gH2AX signaling following various radiation quality exposures. These studies revealed differences in gH2AX foci depending on radiation quality, with the lower energy Fe600 MeV/u, having a greater delta ray component, showing more intense foci as compared to Fe 1000 MeV/u. We then went on to perform further more extensive studies to better understand ATM/TGFb cross-talk in esophageal epithelial cells and proteins involved in the stress response following high vs low doses of radiation. Multiplex studies were performed on ten different stress response phospho-proteins (ATF2, cJun, JNK, MSK,1, p38, STAT1, Hsp27, Erk ½, and MEK1) using an ELISA luminex 96 well format, plus and minus TGFß, its inhibitor, an ATM inhibitor, as well as a combination of both TGFß addition and ATM inhibition. This work enabled us to better define the cross talk in EPC cells with radiation and how radiation dose may differentially affect this response. These studies have been very informative and revealed a subset of proteins with a unique dose-dependent, protein dependent, and TGFß and ATM

dependent response. For example, ATF2 and Hsp27 both show a radiation dose and TGFß dependent increase in expression, whereas MSK1 is decreased with dose and TGFß addition. When both ATM is inhibited and TGFß is added, a radiation dose specific change is noted with ATF2, but effect is only at the lower 0.1 Gy dose. This work has allowed us to better define the cross-talk between ATM and TGFß and how stress response proteins following radiation, are uniquely affected by radiation and influenced by these pathways and the dependence upon dose. The work was initially performed following Xray but in the last year of this study we have also done studies with Fe 600 MeV/u exposure and are currently writing up the findings of these studies, and the unique radiation quality differences noted.

MicroRNAs are a recently discovered class of non-coding RNAs that have been shown to play important roles in gene regulation. They typically act at the transcriptional level to fine tune expression levels. To identify miRNAs changed following high LET exposure, which may be important in influencing cancer risk in esophageal cells, we studied miRNA changes following Si173 MeV/u exposure in 3D organotypic cultures. We have identified a number of miRNA that are upregulated as well as down regulated following these exposures and are currently comparing data with low LET cesium exposure and combining these results with the Luminex studies into a paper.

In total these studies have identified that both TGFß and ATM are critical in esophageal cells responses post radiation, that the pathways exhibit cross-talk, and that this cross talk may be differentially regulated depending on dose. A number of the miRNAs found to be altered in our work have also been noted to be differentially regulated in esophageal cancer and in some cases noted to be serum markers of esophageal cancer, potentially providing biomarkers to better define cancer risk following high LET exposures.

Bibliography Type:	Description: (Last Updated: 11/01/2023)